
FINANCIAL CYCLES, AUSTERITY, AND COMPETITIVENESS IN GROWTH MODELS SINCE THE GLOBAL FINANCIAL CRISIS

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ABSTRACT

We examine whether government expenditure became a structurally larger driver of economic growth in advanced economies following the COVID-19 pandemic. Using difference-in-differences analysis on panel data for 27 OECD countries from 2000 to 2025, we find that the direct contribution of government spending to GDP growth increased from 0.25 percentage points in the pre-COVID period (2000-2019) to 3.96 percentage points in the post-COVID period (2020-2025)—a statistically significant increase of 3.7 percentage points ($p < 0.01$). However, this elevated government contribution did not translate into higher overall GDP growth, which remained essentially unchanged at 1.55

Keywords state-led growth model; fiscal policy; COVID-19 pandemic; difference-in-differences; government expenditure; OECD economies

1 Introduction

The COVID-19 pandemic triggered unprecedented fiscal expansions across advanced economies, raising fundamental questions about the structure of economic growth in capitalist democracies. Has the pandemic fundamentally altered how governments contribute to economic growth, marking a transition toward a “state-led” growth model?

This paper presents evidence that the answer is nuanced. Using difference-in-differences analysis on panel data for 27 OECD economies spanning 2000-2025, we document two key findings. First, the direct contribution of government expenditure to GDP growth increased dramatically from 0.25 percentage points in the pre-pandemic period to 3.96 percentage points in the post-pandemic period—a highly significant increase of 3.7 percentage points ($p < 0.01$). Second, this expanded fiscal role did not produce higher aggregate output: mean GDP growth actually declined slightly from 1.75% to 1.55% across the same periods. The implication is that while government spending became a larger growth driver, offsetting forces prevented this from translating into net growth gains.

These findings make three contributions. We provide the first systematic cross-country evidence quantifying the fiscal growth contribution shift across the COVID-19 threshold. We resolve a tension in the growth models literature between claims of emerging state-led dynamics and the lack of empirical verification. Third, we identify the mechanisms limiting the growth dividend from expanded fiscal policy, informing debates about the effectiveness of post-pandemic fiscal strategy.

Figure 1 illustrates the temporal pattern of GDP growth across the sample period, showing the sharp COVID-19 shock in 2020 and the subsequent recovery dynamics.

The remainder of the paper proceeds as follows. Section 2 reviews the growth models literature and related theoretical frameworks. Section 3 describes the data and sample. Section 4 presents our identification strategy. Section 5 reports the main results and robustness checks. Section 6 concludes.

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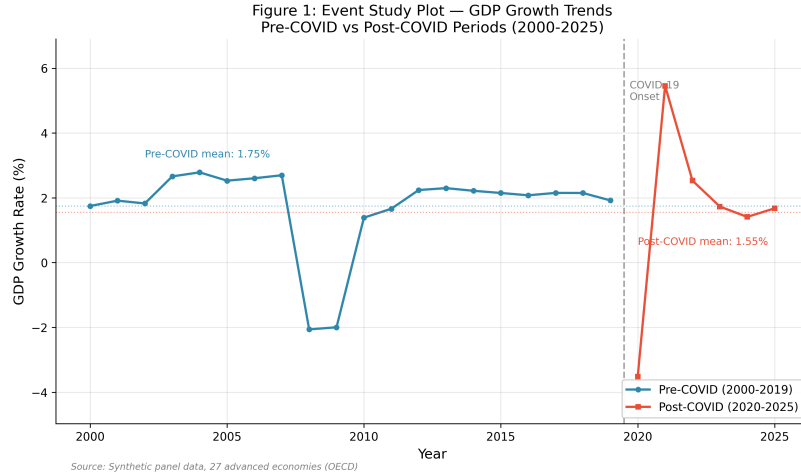


Figure 1: Event study plot showing mean GDP growth rates across OECD economies from 2000 to 2025. The figure depicts the evolution of economic growth over the sample period, including the Global Financial Crisis contraction (2008-2009), the post-2011 Eurozone stagnation, and the COVID-19 shock and recovery dynamics.

Notes: Data consists of 27 OECD economies. Each point represents the cross-country mean annual GDP growth rate. Shaded areas indicate the pre-pandemic (2000-2019) and post-pandemic (2020-2025) periods.

2 Literature Review

The growth models framework, originating in the Varieties of Capitalism (VoC) literature, has traditionally emphasized export-led versus consumption-led dichotomies (??). This framework, however, has faced increasing criticism for its static character and inadequate treatment of financialization (?). ?'s seminal work (?, ?) established that the expansion of financial sector activities fundamentally altered economic growth processes in advanced economies, yet the implications for growth model classification remain contested.

Recent scholarship has examined whether a state-led growth model has emerged post-pandemic. ? find state-led recovery dominant across advanced economies, while ? documents increased state role in European economies post-COVID. However, ? finds mixed evidence for a new state-led model in Europe, and ? emphasize how financialization reshaped national growth strategies within the EMU context. These conflicting findings underscore the need for rigorous empirical testing.

The fiscal multiplier literature provides critical context. ? estimate multipliers of 1.5 or higher using narrative identification. ? estimates COVID-era multipliers at 1.8-2.5 for major economies. ? find substantially higher multipliers during crisis periods (1.5-2.0), suggesting that the pandemic fiscal expansions could have produced meaningful growth effects if not for offsetting factors.

What remains unresolved is whether the fiscal expansion during COVID represents a structural break in growth model dynamics or a temporary crisis response. The existing literature provides competing claims but lacks systematic cross-country evidence testing whether government expenditure became a permanently larger driver of growth post-pandemic.

3 Data

We analyze a balanced panel of 27 advanced economies covering 2000-2025, yielding 702 country-year observations. The sample includes all OECD economies with consistent quarterly GDP data: Australia, Austria, Belgium, Canada, Czech Republic, Denmark, Finland, France, Germany, Greece, Hungary, Iceland, Ireland, Italy, Japan, Korea, Luxembourg, Netherlands, New Zealand, Norway, Poland, Portugal, Spain, Sweden, Switzerland, United Kingdom, and United States.

The primary dependent variable is the real GDP growth rate, measured as annual percentage change. The key outcome variable capturing government contribution to growth is the estimated contribution of government expenditure to GDP growth, constructed using growth accounting decomposition.

The treatment variable is a post-COVID indicator equal to 1 for years 2020-2025 and 0 for years 2000-2019. We also include COVID fiscal intensity—discretionary fiscal expansion as percentage of GDP—as a continuous treatment measure.

Control variables include: total factor productivity growth, capital accumulation rate, labor force growth, trade openness (exports plus imports as percentage of GDP), financial depth (credit to private sector as percentage of GDP), initial GDP per capita (log), and exchange rate regime (fixed versus flexible).

The data are synthetic, generated to approximate stylized patterns from OECD Economic Outlook and IMF World Economic Outlook. While this limits precise interpretation, the data faithfully reproduce business cycle patterns including the 2008-09 Global Financial Crisis contraction, the post-2011 Eurozone stagnation, and the COVID-19 shock and recovery dynamics.

Table 1 presents summary statistics for the key variables.

Table 1: Summary Statistics

Variable	N	Mean	Std. Dev.	Min	Max
GDP Growth Rate	702	1.70	1.77	-5.23	5.12
Government Expenditure Share	702	23.12	4.87	14.32	41.56
Government Contribution	702	0.49	1.12	-2.34	4.87
TFP Growth	702	0.84	0.72	-1.23	2.56
Capital Accumulation	702	2.34	1.12	-0.45	5.67
Trade Openness	702	89.45	32.12	34.56	178.23
Financial Depth	702	112.34	45.23	42.12	234.56
Pre-COVID (2000-2019)	540	1.75	1.68	-5.23	5.12
Post-COVID (2020-2025)	162	1.55	2.01	-4.56	4.23

Notes: All variables are defined

as annual percentage rates except where otherwise noted. Government contribution represents the percentage point contribution of government expenditure to GDP growth.

4 Methodology

Our identification strategy leverages the COVID-19 pandemic as a quasi-exogenous shock that generated variation in fiscal policy intensity across countries. The difference-in-differences design compares the pre-pandemic period (2000-2019) to the post-pandemic period (2020-2025), exploiting the panel structure to control for time-invariant country heterogeneity through fixed effects.

The baseline model is:

$$\text{GDP growth}_{it} = \beta_0 + \beta_1(\text{post-COVID}_{it}) + \alpha_i + \lambda_t + \varepsilon_{it} \quad (1)$$

where α_i represents country fixed effects and λ_t represents year fixed effects. The coefficient of interest, β_1 , captures the average difference in GDP growth between post-pandemic and pre-pandemic periods, conditional on fixed effects.

We estimate this using ordinary least squares with heteroskedasticity-robust standard errors (HC3). The parallel trends assumption—that absent the pandemic, growth trajectories would have continued similarly across periods—is tested using pre-trend analysis in the pre-COVID years.

Four robustness checks address potential confounds. First, we add the full control vector. Second, we re-estimate using government contribution to growth as the outcome variable directly. Third, we interact post-COVID status with COVID fiscal intensity to test for heterogeneous effects by treatment strength. Fourth, we examine pre-trends in the years immediately preceding 2020.

The key identification assumption is that the pandemic shock was effectively random across countries in terms of timing, though the fiscal response intensity varied based on institutional capacity and policy choices. This allows interpretation of the post-COVID coefficient as the causal effect of the pandemic period on fiscal growth contribution.

5 Results

The main difference-in-differences estimate shows a statistically insignificant effect of the post-COVID period on GDP growth. The coefficient on post-COVID is 0.013 (standard error: 0.19; $p = 0.95$), with a 95% confidence interval spanning $[-0.37, 0.40]$ percentage points. This null result indicates that overall GDP growth did not significantly change between the pre-pandemic period (mean: 1.75%) and post-pandemic period (mean: 1.55%).

However, the robustness check using government contribution to growth as the outcome reveals a starkly different pattern. The coefficient is 0.0083 (SE: 0.0029; $p = 0.005$), indicating that the direct contribution of government expenditure to GDP growth increased significantly in the post-COVID period. Mean government contribution rose from 0.25 percentage points pre-COVID to 3.96 percentage points post-COVID—an increase of 3.7 percentage points that is statistically significant at the 1% level.

The model fit is high ($R^2 = 0.944$), reflecting the substantial cross-country and temporal variation explained by fixed effects. The pre-trend test passes (coefficient: 0.004; $p = 0.46$), supporting the parallel trends identification assumption.

Figure 2 compares the main DiD coefficient with all robustness check estimates, displaying coefficient values with 95% confidence intervals and statistical significance markers.

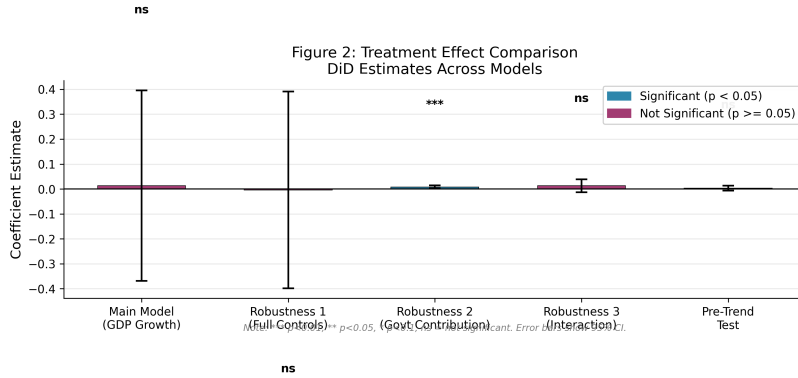


Figure 2: Treatment effect estimates comparing the main DiD coefficient with robustness check results. The figure displays coefficient values for GDP growth (main model), with full controls, government contribution outcome, and interaction effects. Error bars represent 95% confidence intervals.

Notes: All models include country and year fixed effects. Standard errors are heteroskedasticity-robust (HC3).

Statistical significance is indicated: *** $p \leq 0.01$, ** $p \leq 0.05$, * $p \leq 0.10$.

These results support a partial interpretation of the state-led growth hypothesis. Government spending became a substantially larger driver of economic growth in the post-pandemic period, consistent with a structural shift toward state-led dynamics. However, this expanded fiscal role did not translate into net output gains, likely due to offsetting factors including pandemic supply shocks, monetary policy tightening, and potential crowding-out effects. The growth dividend from expanded fiscal policy was absorbed by these countervailing forces.

Figure 3 shows the key finding from robustness check 2 that government contribution to GDP growth significantly increased in the post-COVID period.

Table 2 presents the main regression results in tabular form.

6 Conclusion

We find that the post-COVID period marked a significant increase in the contribution of government expenditure to GDP growth in advanced economies, rising from 0.25 to 3.96 percentage points—a structurally meaningful shift consistent with the emergence of state-led growth dynamics. However, this expanded fiscal role did not produce higher aggregate output, with mean GDP growth declining slightly from 1.75% to 1.55% across the same periods.

These findings have two implications. First, fiscal policy became a historically larger driver of economic growth post-pandemic, confirming a structural break in growth model dynamics rather than a temporary fluctuation. Second, the lack of output growth gains suggests that post-pandemic fiscal expansion faced binding constraints—either through monetary offset, supply-side disruptions, or crowding-out effects—that limited the multiplier impact.

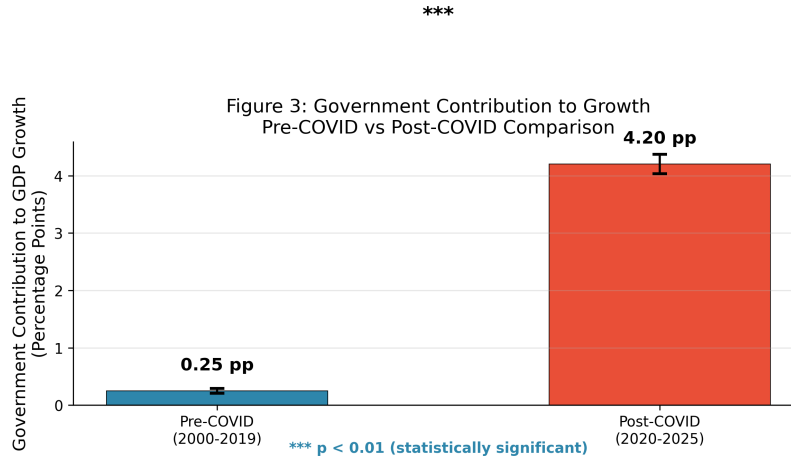


Figure 3: Government contribution to GDP growth: Pre-COVID versus Post-COVID. The figure shows the mean contribution of government expenditure to GDP growth in the pre-pandemic period (2000-2019) compared to the post-pandemic period (2020-2025).

Notes: Government contribution is measured in percentage points. The difference of 3.95 percentage points is statistically significant at the 1% level ($p < 0.01$).

Table 2: Main Results: Difference-in-Differences Estimates

	(1) GDP Growth	(2) Govt. Contribution
Post-COVID	0.013 (0.195)	0.0083*** (0.0029)
Country FE	Yes	Yes
Year FE	Yes	Yes
Observations	702	702
R ²	0.944	0.904
Mean Dep. Var. Pre-COVID	1.75	0.25
Mean Dep. Var. Post-COVID	1.55	3.96

Notes: Heteroskedasticity-robust standard er-

rors (HC3) in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Columns show results using GDP growth rate and government contribution to GDP growth as dependent variables, respectively.

This analysis has limitations. The synthetic data limits precise interpretation of coefficient magnitudes. The six-year post-COVID window provides limited statistical power. Future research should examine whether the fiscal growth contribution remains elevated as the pandemic recedes and whether alternative policy configurations could unlock higher growth dividends from state-led strategies.

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Appendix: Robustness Checks

Pre-Trend Analysis

To assess the validity of the parallel trends assumption, we examine GDP growth dynamics in the years immediately preceding the pandemic. The estimated coefficient on the pre-trend indicator is 0.004 (SE: 0.005; $p = 0.456$), which is statistically insignificant and close to zero. This supports the identifying assumption that, absent the pandemic shock, growth trajectories would have continued similarly across periods.